# Mathematical and Computational Methods

Exercise Sheet 10: Integration II: techniques and applications

This sheet accompanies *Unit 10 (Integration II: techniques and applications)*. It exercises integration by parts, partial fractions and the trigonometric methods, and the applications—areas between curves, volumes of revolution, average values and expectations. All methods are those of Unit 10 and earlier units; no later-unit material is needed. Give exact answers (integers, fractions or simple surds) unless asked to approximate.

**How to use this sheet.** Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

**Solutions.** This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit10}"`) and recompile.

## 1 Warm-up drills

*Short routine problems — at least one per skill, to build fluency. Find the indefinite integral (with $+C$) unless limits are shown.*

### 1.1 Integration by parts

**Exercise 1.1** *($\star$ drill — by parts)*. $\displaystyle\int x\,\mathrm{e}^{2x}\,\mathrm{d}x.$

**Exercise 1.2** *($\star$ drill — by parts)*. $\displaystyle\int x\cos x\,\mathrm{d}x.$

**Exercise 1.3** *($\star$ drill — by parts)*. $\displaystyle\int x\ln x\,\mathrm{d}x.$

### 1.2 Partial fractions

**Exercise 1.4** *($\star$ drill — partial fractions)*. $\displaystyle\int \frac{\mathrm{d}x}{(x-1)(x+1)}.$

**Exercise 1.5** *($\star$ drill — partial fractions)*. $\displaystyle\int \frac{3x+1}{x(x+1)}\,\mathrm{d}x.$

### 1.3 Trigonometric integrals

**Exercise 1.6** *(⋆ drill — trig integral)***.** $\int \sin^2 x \, \mathrm{d}x$.

**Exercise 1.7** *(⋆ drill — trig integral)***.** $\int \cos^3 x \, \mathrm{d}x$.

**Exercise 1.8** *(⋆ drill — trig integral)***.** $\int \tan^2 x \, \mathrm{d}x$.

### 1.4 Trigonometric substitution

**Exercise 1.9** *(⋆ drill — trig substitution)***.** $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{9-x^2}}$.

**Exercise 1.10** *(⋆ drill — standard form)***.** $\displaystyle\int \frac{\mathrm{d}x}{4+x^2}$.

### 1.5 Areas, volumes, averages

**Exercise 1.11** *(⋆ drill — area between curves)***.** Find the area enclosed between $y = 2x$ and $y = x^2$.

**Exercise 1.12** *(⋆ drill — volume of revolution)***.** The region under $y = \sqrt{x}$, $0 \le x \le 4$, is revolved about the $x$-axis. Find the volume.

**Exercise 1.13** *(⋆ drill — average value)***.** Find the average value of $f(x) = x^2$ on $[0, 3]$.

### 1.6 Improper integrals, probability, setting up

**Exercise 1.14** *(⋆ drill — improper integral)***.** $\displaystyle\int_1^\infty \frac{\mathrm{d}x}{x^3}$.

**Exercise 1.15** *(⋆ drill — density)***.** The random variable $X$ has density $f(x) = 3x^2$ on $[0, 1]$ (and 0 elsewhere). Verify it is a density, then find $P(X > \frac{1}{2})$ and $\mathbb{E}[X]$.

**Exercise 1.16** *(⋆ drill — setting up)***.** A rod occupies $0 \le x \le L$ with linear density $\rho(x) = kx$. Find its mass.

## 2 Tutorial problems

*Anchored on the unit's worked examples; many have several parts, so show your method. This bank runs to about two-and-a-half tutorials' worth of material.* A ▶ marks the parts forming the *live core* for a single 2-hour slot (about nine parts, one per technique); work the rest — and all of §3 — at home.

**Exercise 2.1** *(⋆⋆ core — by parts (worked-example anchor))***.** Evaluate

$$\blacktriangleright\ \text{(a)} \int x^2 e^x \, \mathrm{d}x, \quad \text{(b)} \int x^2 \ln x \, \mathrm{d}x, \quad \text{(c)} \int \arctan x \, \mathrm{d}x, \quad \text{(d)} \int e^x \sin x \, \mathrm{d}x.$$

**Exercise 2.2** *(⋆⋆ core — partial fractions)***.** Evaluate

$$\text{(a)} \int \frac{5x-3}{(x-1)(x+1)} \, \mathrm{d}x, \quad \text{(b)} \int \frac{3x+5}{(x+1)^2} \, \mathrm{d}x, \quad \blacktriangleright\ \text{(c)} \int \frac{\mathrm{d}x}{x^2+4x+13}, \quad \text{(d)} \int \frac{x^3}{x^2+1} \, \mathrm{d}x.$$

**Exercise 2.3** *(⋆⋆ core — trigonometric integrals)*. Evaluate

$$\text{(a)} \int \sin^2 x \cos^2 x \, \mathrm{d}x, \quad ▶ \text{(b)} \int_0^{\pi/2} \sin^3 x \cos^2 x \, \mathrm{d}x, \quad \text{(c)} \int \tan^4 x \, \mathrm{d}x.$$

**Exercise 2.4** *(⋆⋆ core — trigonometric substitution)*. Evaluate

$$▶ \text{(a)} \int \sqrt{9 - x^2} \, \mathrm{d}x, \qquad \text{(b)} \int \frac{\mathrm{d}x}{(x^2+4)^{3/2}}.$$

**Exercise 2.5** *(⋆⋆ core — area between curves (worked-example anchor))*.

(a) ▶ Sketch the region first, then find the area enclosed between the parabola $y = x^2$ and the line $y = x + 2$.

(b) Find the total area between $y = \sin x$ and $y = \cos x$ for $0 \le x \le \frac{\pi}{2}$.

**Exercise 2.6** *(⋆⋆ core — volumes of revolution)*.

(a) ▶ Find the volume when the region under $y = \sin x$, $0 \le x \le \pi$, is revolved about the $x$-axis.

(b) A right circular cone has base radius $r$ and height $h$. By revolving the line $y = \frac{r}{h}x$, $0 \le x \le h$, about the $x$-axis, derive $V = \frac{1}{3}\pi r^2 h$.

**Exercise 2.7** *(⋆⋆ core — average value)*.

(a) ▶ Find the average value of $f(x) = 3x^2$ on $[0, 2]$.

(b) *(Physics.)* A particle moves with velocity $v(t) = 6t - t^2$ $(\mathrm{m\,s^{-1}})$ for $0 \le t \le 6$. Find its average velocity over this interval.

**Exercise 2.8** *(⋆⋆ core — improper integrals)*. Evaluate, or show divergent:

$$\text{(a)} \int_0^\infty \mathrm{e}^{-2x} \, \mathrm{d}x, \quad ▶ \text{(b)} \int_0^\infty x \, \mathrm{e}^{-x} \, \mathrm{d}x, \quad \text{(c)} \int_1^\infty \frac{\mathrm{d}x}{x^p} \quad (p > 0).$$

**Exercise 2.9** *(⋆⋆ core — probability and expectation (data science))*. *(Exponential law.)* The lifetime $T$ (hours) of a component has density $f(t) = \lambda \mathrm{e}^{-\lambda t}$ on $[0, \infty)$ with $\lambda = \frac{1}{2}$.

(a) ▶ Find $P(T > 2)$.

(b) Find $P(1 < T < 3)$.

(c) Find the mean lifetime $\mathbb{E}[T]$.

**Exercise 2.10** *(⋆⋆ core — setting up (physics, work))*. A uniform rope of length $L = 10\,\mathrm{m}$ and mass $\mu = 2\,\mathrm{kg}$ per metre hangs from a winch at the top of a building. Taking $g = 9.8\,\mathrm{m\,s^{-2}}$, find the work done to wind the whole rope to the top.

**Exercise 2.11** *(⋆⋆⋆ challenge — volume by cross-sections)*. A solid has a circular base of radius $R$ in the $xy$-plane. Every cross-section perpendicular to a fixed diameter is a square. Find its volume.

**Exercise 2.12** *(▶ ⋆⋆ core — spot the error (volume of revolution))*. A student finds the volume generated by revolving $y = \sqrt{x}$, $0 \le x \le 4$, about the $x$-axis and writes

$$V = \pi \int_0^4 \sqrt{x} \, \mathrm{d}x = \pi \left[ \tfrac{2}{3} x^{3/2} \right]_0^4 = \frac{16\pi}{3}.$$

The genuine volume (Warm-up §1) is $8\pi$. What went wrong, and what is the correct set-up?

# 3 Further practice (home)

*A home bank, routine to harder. Challenge / extension items are marked ⋆⋆⋆ challenge. The last group is conceptual — true/false and "spot the error".*

## 3.1 Integration by parts

**Exercise 3.1** (*⋆⋆ core — by parts (twice)*). $\int x^2 \cos x \, \mathrm{d}x$.

**Exercise 3.2** (*⋆⋆ core — by parts (twice)*). $\int (\ln x)^2 \, \mathrm{d}x$.

**Exercise 3.3** (*⋆⋆ core — cyclic by parts*). $\int \mathrm{e}^{2x} \sin 3x \, \mathrm{d}x$.

**Exercise 3.4** (*⋆⋆ core — by parts*). $\int x \arctan x \, \mathrm{d}x$.

**Exercise 3.5** (*⋆⋆ core — definite by parts*). $\int_0^1 x \, \mathrm{e}^{-x} \, \mathrm{d}x$.

## 3.2 Partial fractions

**Exercise 3.6** (*⋆ drill — partial fractions*). $\int \frac{\mathrm{d}x}{x^2 - 9}$.

**Exercise 3.7** (*⋆⋆ core — repeated factor*). $\int \frac{x}{(x-1)^2} \, \mathrm{d}x$.

**Exercise 3.8** (*⋆⋆ core — three linear factors*). $\int \frac{\mathrm{d}x}{x(x-1)(x+1)}$.

**Exercise 3.9** (*⋆⋆ core — linear over quadratic*). $\int \frac{2x+3}{x^2+9} \, \mathrm{d}x$.

**Exercise 3.10** (*⋆⋆ core — improper fraction*). $\int \frac{x^2}{x^2+4} \, \mathrm{d}x$.

## 3.3 Trigonometric integrals and substitution

**Exercise 3.11** (*⋆⋆ core — trig integral*). $\int \sin^4 x \, \mathrm{d}x$.

**Exercise 3.12** (*⋆⋆ core — trig integral*). $\int \cos^5 x \, \mathrm{d}x$.

**Exercise 3.13** (*⋆⋆ core — secant power*). $\int \sec^4 x \, \mathrm{d}x$.

**Exercise 3.14** (*⋆ drill — definite trig integral*). $\int_0^{\pi/4} \tan^2 x \, \mathrm{d}x$.

**Exercise 3.15** (*⋆⋆ core — trig substitution*). $\int \sqrt{1 - x^2} \, \mathrm{d}x$.

**Exercise 3.16** (*⋆⋆⋆ challenge — trig substitution*). $\int \frac{\mathrm{d}x}{x^2\sqrt{4 - x^2}}$.

**Exercise 3.17** *(⋆⋆ core — trig substitution (secant)).* $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{x^2-9}}$ (for $x > 3$).

## 3.4 Areas, volumes, averages

**Exercise 3.18** *(⋆ drill — volume of revolution (sketch first)).* Sketch the region under $y = \dfrac{1}{x}$ for $1 \le x \le 2$, then find the volume of the solid obtained by revolving that region about the $x$-axis.

**Exercise 3.19** *(⋆⋆ core — area between curves).* Find the area enclosed between $y = x^2$ and $y = 8 - x^2$.

**Exercise 3.20** *(⋆⋆ core — volume, washer extension).* *(Extension.)* The region between $y = x$ and $y = x^2$ $(0 \le x \le 1)$ is revolved about the $x$-axis. Treating the solid as the difference of two discs, find its volume.

**Exercise 3.21** *(⋆ ⋆ ⋆ challenge — volume by cross-sections).* A solid has base the disc $x^2 + y^2 \le 1$. Cross-sections perpendicular to the $x$-axis are equilateral triangles. Find its volume.

**Exercise 3.22** *(⋆⋆ core — average value).* Find the average value of $f(x) = \sin x$ over $[0, \pi]$, and of $g(x) = \frac{1}{x}$ over $[1, \mathrm{e}]$.

## 3.5 Improper integrals

**Exercise 3.23** *(⋆ drill — substitution + improper).* $\displaystyle\int_0^\infty x\,\mathrm{e}^{-x^2}\,\mathrm{d}x$.

**Exercise 3.24** *(⋆⋆ core — partial fractions + improper).* $\displaystyle\int_2^\infty \frac{\mathrm{d}x}{x^2-1}$.

**Exercise 3.25** *(⋆ ⋆ ⋆ challenge — by parts + improper).* $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\,\mathrm{d}x$.

## 3.6 Probability and expectation

**Exercise 3.26** *(⋆⋆ core — bounded density).* $X$ has density $f(x) = \frac{3}{2}(1 - x^2)$ on $[0, 1]$. Verify it is a density, then find $P(X < \frac{1}{2})$ and $\mathbb{E}[X]$.

**Exercise 3.27** *(⋆⋆ core — find the constant).* Find $k$ so that $f(x) = kx$ is a probability density on $[0, 4]$, then compute $\mathbb{E}[X]$.

**Exercise 3.28** *(⋆⋆ core — exponential law, median).* A lifetime has exponential density with mean 5 (so $\lambda = \frac{1}{5}$). Find $P(X > 10)$ and the median $m$ (the value with $P(X \le m) = \frac{1}{2}$).

**Exercise 3.29** *(⋆ ⋆ ⋆ challenge — general result).* For the uniform density $f(x) = \dfrac{1}{b-a}$ on $[a, b]$, show that $\mathbb{E}[X] = \dfrac{a+b}{2}$.

## 3.7 Setting up integrals

**Exercise 3.30** *(⋆⋆ core — work, Hooke's law (physics)).* A spring obeys Hooke's law with stiffness $k = 200\,\mathrm{N\,m^{-1}}$, so the force to hold it stretched by $x$ metres is $F(x) = kx$. Find the work done in stretching it from its natural length to $x = 0.1\,\mathrm{m}$.

**Exercise 3.31** *($\star\star\star$ challenge — mass and centre of mass)*. A rod on $0 \le x \le L$ has density $\rho(x) = \rho_0\left(1 + \frac{x^2}{L^2}\right)$. Find its mass $M$ and the position $\bar{x} = \dfrac{1}{M}\int_0^L x\,\rho(x)\,\mathrm{d}x$ of its centre of mass.

## 3.8 Conceptual: true/false and spot-the-error

**Exercise 3.32** *($\star\star$ core — why we need numerical integration)*. Explain briefly why $\displaystyle\int_0^1 \mathrm{e}^{-x^2}\,\mathrm{d}x$ cannot be evaluated by finding an elementary antiderivative, and describe how you would instead obtain a numerical value for it.

**Exercise 3.33** *($\star\star$ core — true or false?)*. Decide, with a one-line reason, whether each statement is true or false.

(a) Every rational function has an elementary antiderivative (after polynomial division, if needed).

(b) $\displaystyle\int_{-\infty}^{\infty} x\,\mathrm{d}x = 0$ because the integrand is odd.

(c) The average value of a continuous $f$ on $[a, b]$ always lies between the minimum and maximum of $f$ on $[a, b]$.

(d) The volume obtained by revolving $y = f(x)$ about the $x$-axis is $\pi\int_a^b f(x)\,\mathrm{d}x$.

**Exercise 3.34** *($\star$ drill — spot the error)*. A student writes $\displaystyle\int \frac{\mathrm{d}x}{x^2+4} = \arctan\frac{x}{2} + C$. What is wrong, and what is the correct answer?

**Exercise 3.35** *($\star\star$ core — spot the error)*. A student computes $\displaystyle\int_{-1}^{1}\frac{\mathrm{d}x}{x^2} = \left[-\frac{1}{x}\right]_{-1}^{1} = -1-1 = -2.$ Why is this nonsense, and what is the correct conclusion?

## 3.9 Challenges and extensions

**Exercise 3.36** *($\star\star\star$ challenge — reduction formula)*. Show, using integration by parts, that for $n \ge 2$

$$\int \sin^n x\,\mathrm{d}x = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x\,\mathrm{d}x,$$

and use it to evaluate $\displaystyle\int_0^{\pi/2}\sin^4 x\,\mathrm{d}x$.

**Exercise 3.37** *($\star\star\star$ challenge — the factorial integral)*. Let $I_n = \displaystyle\int_0^\infty x^n\mathrm{e}^{-x}\,\mathrm{d}x$ for integer $n \ge 0$. Show $I_0 = 1$ and $I_n = n\,I_{n-1}$, and deduce $I_n = n!$.

**Exercise 3.38** *($\star\star\star$ challenge — normalising the Gaussian (data science))*. Given the standard result $\displaystyle\int_{-\infty}^{\infty}\mathrm{e}^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$, find the constant $c$ making $f(x) = c\,\mathrm{e}^{-x^2}$ a probability density on $\mathbb{R}$, and find $\mathbb{E}[X]$.